

For each question, no points will be given without an explanation.

1 Eclat (6 pts)

	Tid	Transaction
	1	B D
	2	B E
DB ₁ =	3	A B C D
	4	A C D
	5	A B C
	6	C D E

In the following questions, we simulate an execution of Eclat algorithm to compute all itemsets X with $\text{supp}(X) \geq 2$ in DB_1 . We assume the order on the items used by Eclat is alphabetical order (i.e., the first projected database is $\text{DB}_1[A]$). The following questions are about this execution of Eclat.

Question 1: Give the representation of the database DB_1 used by Eclat.

Question 2: Give the Tid lists computed by Eclat to represent $\text{DB}_1[A]$. How are they computed?

Question 3: Then Eclat does a recursive call to $\text{MineFrequent}(\text{DB}_1[A])$. What projected databases are computed by Eclat in this recursive call ?

Question 4: What are the frequent itemsets in $\text{DB}_1[A]$? (It is not necessary to simulate Eclat, just give the result.) Explain how to compute the corresponding frequent itemsets in DB_1 .

Question 5: Give the Tid lists that represent $\text{DB}_1[B]$ and $\text{DB}_1[BC]$ computed by Eclat latter in this execution.

2 Constraints (4 pts)

We assume items in a database can be of several types. To simplify, we imagine that there are only two types. $\text{type}(I)$ is the type of an item I . The type of items A, B and C is "1" and the type of items d to h is "2".

We are given the following database:

Tid	A	B	C	d	e	f	g	h
1	0	0	0	0	0	0	1	1
2	0	1	1	1	1	1	1	0
3	0	0	0	1	1	0	1	1
4	1	1	1	1	0	1	1	1
5	0	1	0	1	0	1	0	0
6	0	0	1	0	1	0	0	0

We want to compute the itemsets satisfying the two constraints: $C_s(X) = (\text{supp}(X) \geq 3)$ and $C_{\text{diff}}(X) = (\exists a, b \in X, \text{type}(a) \neq \text{type}(b))$. For instance, $C_{\text{diff}}(ABf)$ and $C_{\text{diff}}(Cg)$ are true and $C_{\text{diff}}(ABC)$ is false.

Question 6: Give a proof that C_{diff} is monotonic.

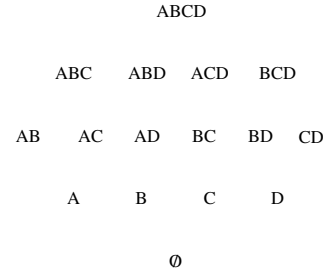
Question 7: how can we use these constraints to remove some of the rows/columns of the database? Which row(s) and which column(s) can be removed in the database and why (detail the order in which they are removed)?

Question 8: Finally, in this database, what are the itemsets that satisfy both constraints?

3 Apriori and Closed Itemsets (3 pts)

$$DB_2 = \begin{array}{c|cccc} & A & B & C & D \\ \hline 1 & 1 & & 1 & 1 \\ 2 & & 1 & & 1 \\ 3 & & 1 & & 1 \\ 4 & 1 & & & \end{array}$$

Lattice of itemsets :



Question 9: Do a simulation of Apriori algorithm on DB_2 . For each iteration of the algorithm, give the set of candidate itemsets C_i , their supports and the set F_i of frequent itemsets.

Question 10: What are the equivalence classes and the closed itemsets in DB_2 ? (Two itemsets are equivalent if they have the same support set). (You can draw the equivalence classes on the itemset lattice.)

4 Krimp (4 pts)

In the code tables below, only the size of the codes of the itemsets are given (in bits), not the codes themselves (since we are not interested in the codes).

$$CT_1 = \begin{array}{|c|c|} \hline \text{Itemset} & \text{Code Size} \\ \hline ABC & 1.5 \\ BC & 2 \\ CD & 2.5 \\ A & 2.5 \\ D & 3 \\ C & 3.5 \\ \hline \end{array}$$

$$DB_3 = \begin{array}{|c|c|} \hline \text{Tid} & \text{Transaction} \\ \hline 1 & ABC \\ 2 & ABC \\ 3 & ABC \\ 4 & BC \\ 5 & BCD \\ 6 & CD \\ \hline \end{array}$$

$$ST = \begin{array}{|c|c|} \hline \text{Itemset} & \text{Code Size} \\ \hline C & 1.4 \\ B & 1.7 \\ A & 2.4 \\ D & 3 \\ \hline \end{array}$$

Question 11: (This question is independent of the two others) In this question, we use Code Table CT_1 . For each of the following transactions, explain which itemsets are used to compress it and what is the size of the compressed transaction:

$T_1 = ABCD$, $T_2 = BCD$, $T_3 = ACD$.

Question 12: Suppose that a code table CT_2 contains itemsets A, B, C, D, CD, ABC. This code table is used to compress DB_3 . In which order do the itemsets appear in the code table? Compute the code sizes of each of these itemsets in CT_2 (if an itemset has a usage of 0, it is removed from the code table).

Recall that $\log_2(ab) = \log_2(a) + \log_2(b)$ and thus $\log_2(a/b) = \log_2(a) - \log_2(b)$.

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\log_2(x)$	0	1	1.6	2	2.3	2.6	2.8	3	3.2	3.3	3.5	3.6	3.7	3.8	3.9	4

Question 13: Given the Standard code Table ST , compute the length of CT_2 .

5 Stream (3 pts)

We consider the following stream:

timestep	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
item	a	c	c	d	a	b	c	b	b	a	a	c	b	b	a	b	a

We want to compute k -icebergs with $k = 3$ (i.e., items with a relative frequency $> 1/3$).

Question 14: Simulate the Martin Flageolet algorithm. For each time step in the stream, give the content (item and count) of each of the 2 registers.

Question 15: Gives the list of 3-icebergs at timestep = 8 (included) and then at timestep = 17 (included).